

CUSTOMER CHURN ANALYSIS

Submitted By:

Name:

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Internship:

Future Interns – Data Analytics Using Python

Task:

Task 2 – Customer Churn Analysis

Objective:

The objective of this project is to analyze customer churn data using Python and identify patterns that influence customer retention. The analysis helps understand customer behavior and supports business decisions to reduce churn.

Tools Used:

Python

VS Code

Pandas

NumPy

Matplotlib

Dataset

Dataset Name: Telco Customer Churn Dataset

Number of Records: 7043

Number of Columns: 21

Data Cleaning:

The following preprocessing steps were performed:

Loaded the dataset using Pandas.

Displayed the first five rows of the dataset.

Checked the dataset shape and column names.

Verified data types using `info()`.

Checked for missing values.

Checked and removed duplicate records (if any).

Converted the TotalCharges column to numeric format.

Filled missing values in TotalCharges using the median value.

Exploratory Data Analysis (EDA)

The following analyses were performed:

Customer Churn Count

Monthly Charges Distribution

Tenure Distribution

Churn by Gender

Churn by Senior Citizen

Visualizations:

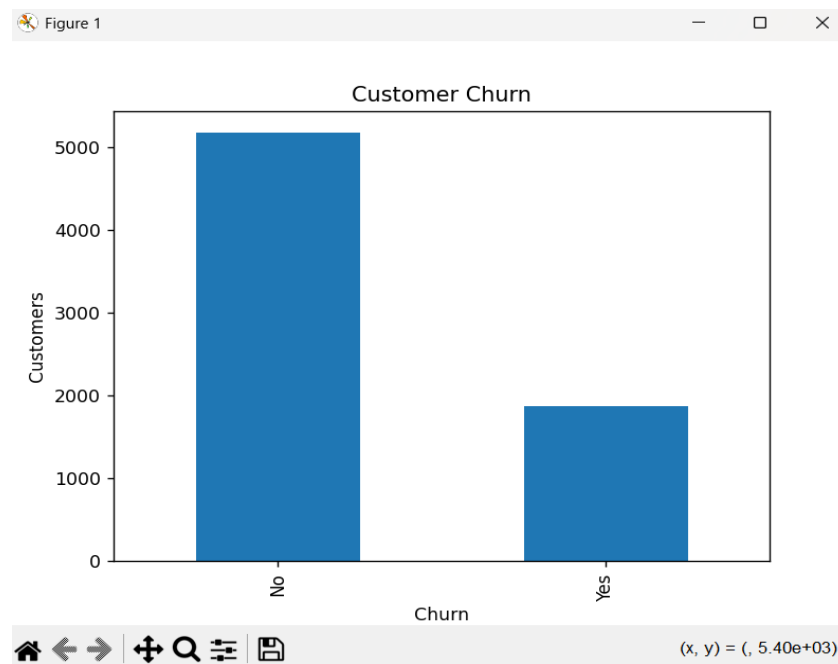


Figure 1: Customer Churn Count

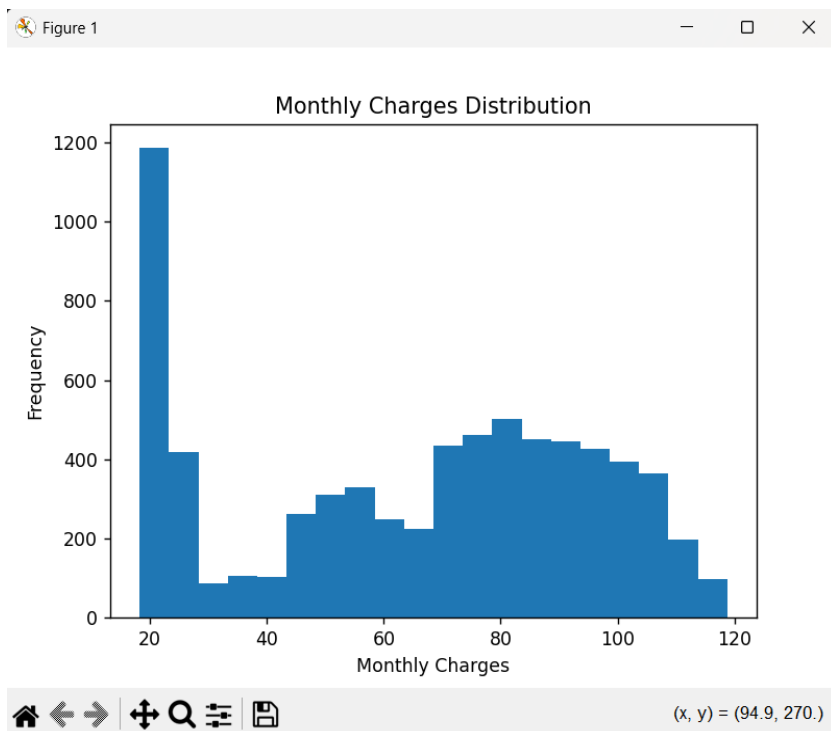


Figure 2: Monthly Charges Distribution

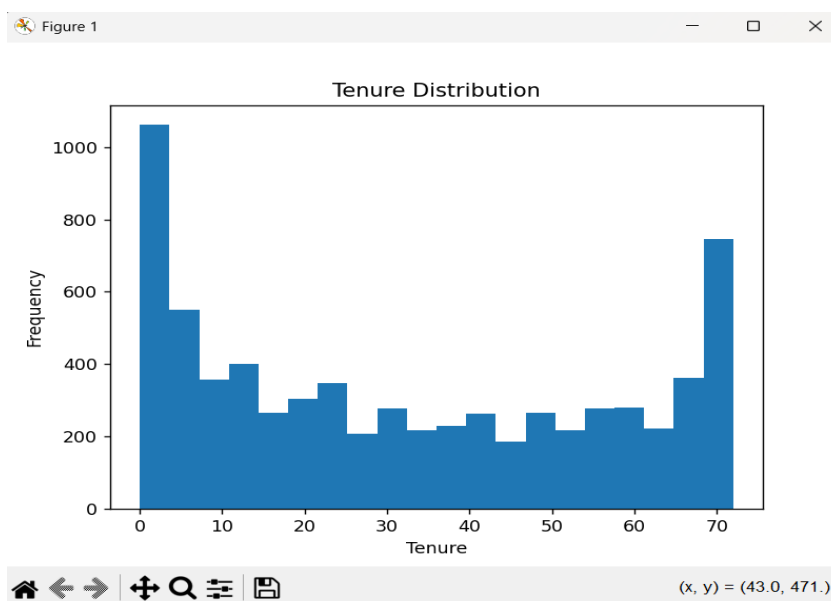


Figure 3: Tenure Distribution

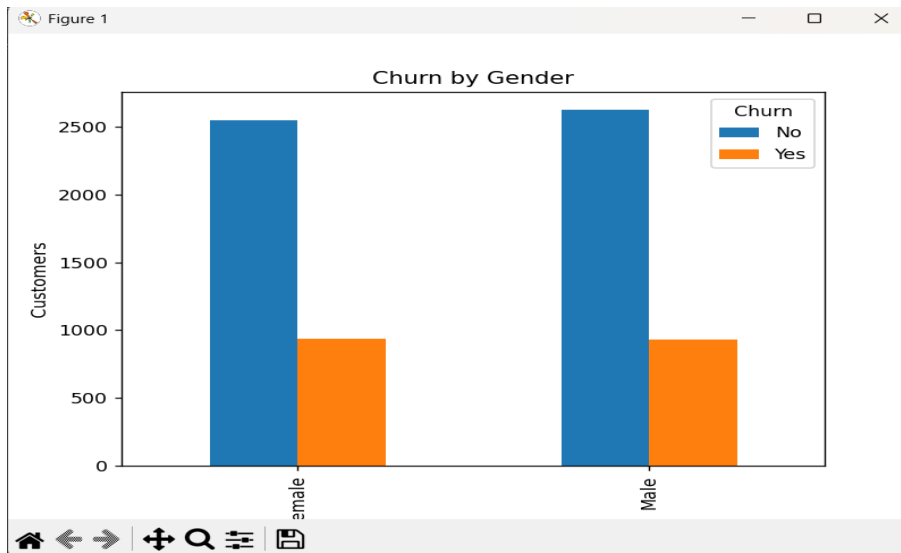


Figure 4: Churn by Gender

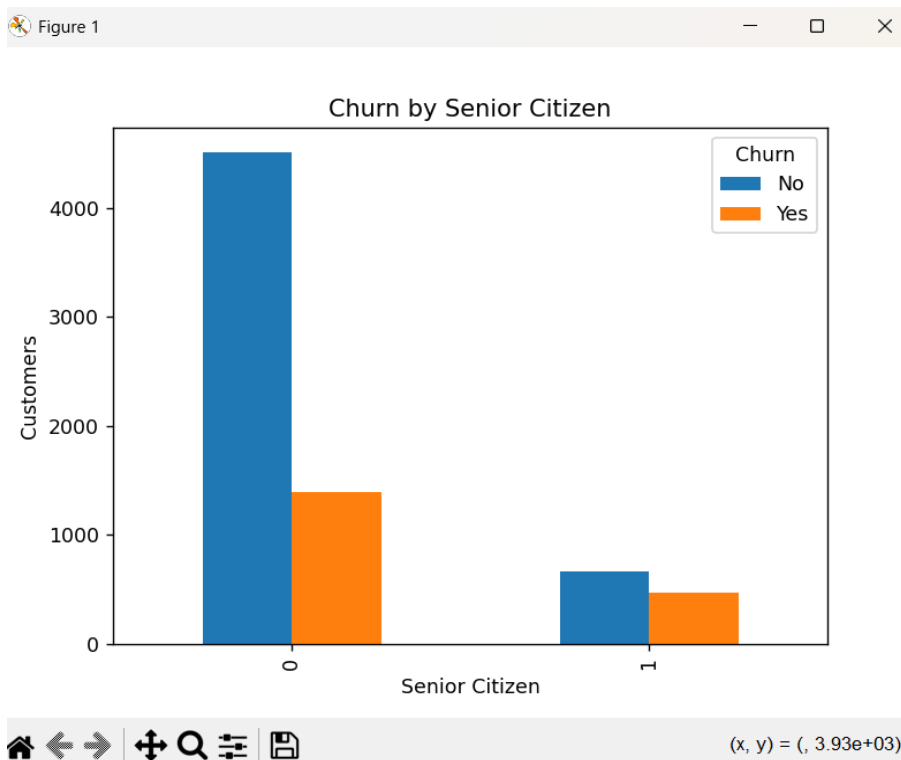


Figure 5: Churn by Senior Citizen

Observations:

The dataset contains 7043 customers and 21 columns.

Most customers did not churn.

Customers with higher monthly charges tend to show more churn.

Customers with shorter tenure are more likely to churn.

Gender has only a small impact on customer churn.

Senior citizens have a relatively higher churn rate compared to non-senior citizens.

Conclusion

Customer churn analysis helps businesses identify customers who are likely to leave the service. By understanding the factors affecting churn, companies can improve customer satisfaction and retention through better plans, offers, and support.